

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. STA/STB-503: Programming with C

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) With the help of examples, explain the following types of constants in C: numeric, character, real, and string constants. [7]
- (b) Write the output of the following C expressions, given `int a=2, b=5, c=3`:
- (i) `m = 2 - a + 5 * 2 / 8 % 3`
 - (ii) `x = a * b / a * b - c`
 - (iii) `w = (a * (c + b)) * (b + 5 - 2)`
 - (iv) `x = b - b / 2 + c`

[7]

Question 2.

- (a) With the help of a program, explain the use of the increment (`++`), decrement (`-`), and conditional (`?:`) operators in C. [7]
- (b) Describe any two input and output functions used in C with examples. [7]

Question 3.

- (a) Describe the `if-else` statement in C. Write a C program to find the maximum of n given numbers. [7]
- (b) With the help of a program, explain the use of the `switch` statement in C. [7]

Question 4.

- (a) With the help of a program, explain the use of `break`, `continue`, and `goto` statements in C. [7]

- (b) Define one-, two-, and multi-dimensional arrays in C. Write a C program to add two matrices. [7]
-

Question 5.

- (a) With the help of examples, explain the use of automatic, external, and static variables in C. [7]
- (b) With the help of a user-defined function, write a C program to find the variance of n given numbers. [7]
-

Question 6.

- (a) With the help of a program, explain the difference between *call by value* and *call by reference* in C. [7]
- (b) Explain the use of pointers in C. Also explain the use of pointers in the implementation of arrays with a suitable example. [7]
-

Question 7.

- (a) Define structure in C. Write a C program demonstrating the use of an array of structures. [7]
- (b) Describe the similarities and differences between structures and unions in C. Write a program demonstrating the use of a union. [7]
-

Question 8. Write short notes on any **two** of the following: [14]

- (a) History of the C programming language
- (b) Recursion in functions
- (c) Use of files in C
- (d) Scope and lifetime of variables
-

[Click here to visit our website](#)
[Click here to visit our contributor](#)
[Click here to visit our contributor](#)
